Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Used Paper Aug-2024

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Give C band and S band frequency range.	[2]
b)	Define modulation and mention different analog modulations.	[3]
c)	Give the equation of AM system in time domain and frequency domain.	[2]
d)	Draw the PAM signal.	[3]
e)	Name different parts of telephone handset.	[2]
f)	Draw typical diagram of token ring LAN.	[3]
g)	Define Total Internal Reflection in optical fiber.	[2]
h)	What are the advantages and disadvantages in positioning satellite in lower orbit?	[3]
i)	Name different types of fading mechanisms in mobile radio environment.	[2]
j)	State the standard frequency bands allocated for mobile communications.	[3]

PART – B

(50 Marks)

2.a) Explain the advantages of modulation.
b) What is mean by frequency translator? Discuss it. [4+6]

OR

3.a) Give different bands of electromagnetic spectrum and their ranges.
b) A two stage amplifier has gain of 150 for first stage and 100 for second stage. Express overall gain in dB. [7+3]

4.a) Explain the method of generation and detection of PAM signals with neat schematics.
b) A band limited signal has a bandwidth equal to 4 KHZ. What sampling rate should be used to guarantee a guard band of 1200 Hz? [6+4]

OR

5.a) Draw the constellation diagram for M-ary PSK System.
b) Explain BPSK system transmitter and receiver with the help of diagrams. [4+6]

6.a) Explain the principles of Cross bar switching in Telecommunication.
b) Given that MTBF=2000 Hrs and MTTR=4 Hrs. Calculate the unavailability for single and dual processor systems for 10 years and 20 years. [6+4]

OR

7.a) Explain Routing Control in Packet Switching.
b) What are the IEEELAN standards? List them. [5+5]

8.a) Explain the Frequency allocations for satellite services.
b) Write short notes on:
i) EIRP
ii) Carrier to Noise Density Ratio. [4+6]

OR

9.a) What are the losses occurred in optical fibers? Explain briefly.
b) Describe with diagram optical receiver with light emitting diode. [5+5]

10.a) Explain the procedure of cell-splitting and advantages of cell sectorization.
b) Describe the RFID communication and its advantages. [5+5]

OR

11.a) Explain the following in a cellular environment.
i) Hand-off mechanism
ii) Cell splitting.
b) Discuss the concept of Bluetooth networks. [5+5]

---ooOoo---

Used Paper Aug-2024